

Joint-tuple counting moves the window-13 excess3 maximum into developed chaos

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25 September 2026

DOI: 10.5281/zenodo.22970080

Version 2 of 10.5281/zenodo.22945800

Abstract

Version 1 of this note reported that, on four diffusively coupled logistic maps, the hybrid excess3 = $0.6\text{Syn} + 0.4\text{Surp}$ is largest near period-doubling accumulation. That location was produced by nested-recd 0.2.2, which flattened joint tuples when it counted the entropies inside Syn. This version keeps the generator, the seeds, the grid, and the weights, and counts joint tuples (nested-recd 0.2.3). On windows of 13 symbols the six-seed mean is largest at the right edge of the grid, $r = 3.95$, where excess3 = 2.897 ± 0.006 . At $r = 3.5675$ the mean is 2.383 ± 0.0004 . Surprise still peaks in the accumulation band, 5.936 ± 0.004 , on 4.11 ± 0.02 joint tuples, and the full-block residual there remains 0.0003 ± 0.0001 . Synergy is 0.0137 ± 0.0021 at $r = 3.5675$ and 1.782 ± 0.010 at $r = 3.95$. The short-window maximum is therefore the chaotic rise of Syn. Leave-one-seed maxima all fall at $r = 3.95$. For windows of length 78 and longer the argmax returns to the accumulation band. At $r = 3.85$ the full-block residual remains 0.192 ± 0.002 . The legacy estimator reproduces every archived window-13 mean of version 1.

Keywords: excess3; Bandt–Pompe patterns; Feigenbaum accumulation; pairwise residual; coupled logistic maps

Resumen. La versión 1 de esta nota situaba el máximo de excess3 = $0.6\text{Syn} + 0.4\text{Surp}$ cerca de la acumulación. Ese lugar salía de nested-recd 0.2.2, que aplanaba las tuplas conjuntas al contar las entropías de Syn. Esta versión conserva el generador, las semillas, la rejilla y los pesos, y cuenta tuplas (nested-recd 0.2.3). En ventanas de 13 símbolos el máximo de las seis semillas está en $r = 3.95$ y vale 2.897 ± 0.006 . En $r = 3.5675$ vale 2.383 ± 0.0004 . La sorpresa sigue haciendo pico en la banda de acumulación (5.936 ± 0.004) sobre 4.11 ± 0.02 tuplas, y el residuo del bloque completo sigue en 0.0003 ± 0.0001 . La sinergia vale 0.0137 ± 0.0021 ahí y 1.782 ± 0.010 en $r = 3.95$. El máximo de la ventana corta lo lleva Syn en el caos. A partir de ventanas de longitud 78 el máximo vuelve a la banda de acumulación. En $r = 3.85$ el residuo del bloque sigue en 0.192 ± 0.002 . El estimador antiguo reproduce cada media de la versión 1.

1 Introduction

A critical transition is often described through the growth of variance and lag-1 autocorrelation [Scheffer et al., 2009]. An ordinal reading asks a different question: whether the joint order of several series is being reorganized [Bandt and Pompe, 2002, Padilla-Villanueva, 2026b]. In that reading, Systemic Tau summarizes pairwise rank concordance, and a discrete clock is built from nested ordinal conjunctions. The third level of the clock is the residual of the joint ordinal law beyond pairwise margins. The hybrid excess3 was specified so that a short window could be scored when that residual is costly to compute [Padilla-Villanueva, 2026a,c]. The weights 0.6 and 0.4 are a declared design choice. They were not fitted to the logistic suite.

On the published cascade of diffusively coupled logistic maps, the pooled-count proxy is higher at $r = 3.85$ than at $r = 3.30$, and its grid maximum lies near the accumulation point, at about 2.38 [Padilla-Villanueva, 2026b]. Version 1 of this note reproduced that maximum and read it as a low-diversity orbit. The reading of the support is kept. The location of the hybrid maximum is not: it was an output of the flattened count inside Syn.

The decomposition uses the published generator, the published seeds, and the published weights, on a grid refined through the accumulation band. Joint tuples are counted with nested-recd 0.2.3. Three comparisons stay fixed. The first splits the hybrid into Syn and Surp. The second asks whether the pairwise maximum-entropy law already produces the score. The third asks how the ordering of stations changes when the window is lengthened, and whether a period-3 window follows the chaotic stations. Res_{pair} is reported beside the proxy.

2 Estimators

Let $S_t \in \{0, \dots, 5\}^4$ be the joint Bandt–Pompe symbol of order $m = 3$ and delay 1 at time t [Bandt and Pompe, 2002]. In a window of length w , write p for the empirical law of the joint symbol and p_i for its marginals. Total correlation, in bits, is

$$\text{TC} = \sum_{i=1}^4 H(p_i) - H(p). \quad (1)$$

The pairwise summary used by the proxy is $(4 - 1)$ times the mean pairwise mutual information. Synergy in this construction is the nonnegative excess of total correlation over that summary,

$$\text{Syn} = [\text{TC} - (d - 1) \bar{I}_{\text{pair}}]_+, \quad d = 4. \quad (2)$$

Surprise retains the positive part of the log-ratio against independence,

$$\text{Surp} = \sum_s p(s) [\log_2 p(s) - \log_2 p_{\text{ind}}(s)]_+, \quad (3)$$

with $p_{\text{ind}}(s) = \prod_i p_i(s_i)$ and the marginal probability floored at 10^{-9} in the published implementation. The hybrid is

$$\text{excess3} = 0.6 \text{Syn} + 0.4 \text{Surp}. \quad (4)$$

These names are local to the proxy. They are not the synergy and redundancy atoms of a partial information decomposition [Williams and Beer, 2010].

The strong residual is a different functional. Let $P^{(2)}$ be the maximum-entropy law with the same pairwise margins as p , obtained by iterative proportional fitting. Then

$$\text{Res}_{\text{pair}} = D_{\text{KL}}(p \parallel P^{(2)}), \quad (5)$$

in bits. A positive value of `excess3` does not imply a positive value of Res_{pair} . Surprise is large for a joint that is far from independence, including a joint that pairs already explain.

3 Protocol

The system is the diffusively coupled logistic family used for the published cascade [Kaneko, 1992, Padilla-Villanueva, 2026b]. Four coordinates obey

$$\begin{aligned} f_i(t) &= r x_i(t)(1 - x_i(t)), \\ x_i(t+1) &= (1 - \varepsilon) f_i(t) + \varepsilon \bar{f}_{-i}(t) + \xi_i(t), \end{aligned} \quad (6)$$

with the mean $\bar{f}_{-i}$ taken over the other three logistic images, $\varepsilon = 0.05$, and $\xi_i(t)$ independent Gaussian noise of standard deviation 0.003. Each coordinate is then clipped to $[0, 1]$. The initial condition is uniform on $[0.2, 0.8]$. Four hundred steps are discarded and 2200 steps are retained. The seed of realization k at parameter r is

$$10000 + \lfloor 10000 r \rfloor + 17k, \quad k = 0, \dots, 5. \quad (7)$$

After the embedding, each series contains 2198 joint symbols.

The parameter grid is the 32-point cascade grid, union a uniform grid of 19 values on $[3.545, 3.590]$, restricted to $[3.05, 3.98]$. Forty-nine distinct values remain. The conventional accumulation value of the logistic family is $r_\infty = 3.5699456$ [Feigenbaum, 1978, 1983]. The nearest grid point is $r = 3.569946$. The period-3 window is represented by $r = 3.828427$.

For windows of length 13 the hybrid is the mean over every consecutive window, which is the published abundance. For every other length in $\{6, 8, 10, 13, 20, 26, 40, 52, 78, 104, 156, 208, 312\}$ the same mean is taken on about 160 evenly spaced windows, the stride being the smallest integer that yields at most that count. The full-block scores use all 2198 symbols as one window. Means and standard errors below are taken across the six seeds. The standard error uses the sample standard deviation with divisor 5.

Two Monte Carlo controls use the invariant joint of each series. Eighty sequences of length 13 are drawn independently from p , and eighty from $P^{(2)}$, with the random generator seeded by the series seed plus 901. A local control repeats the comparison inside 36 evenly spaced windows of length 13, with four draws from that window's own p and four from its pairwise projection. The film strip in Figure 2 is one realization per station: the seed whose mean number of distinct 13-symbol tuples is closest to the six-seed mean, and, inside that series, a run of 16 symbols whose support is closest to the series mean.

With `legacy_pooled_counting` the window-13 means at $r = 3.30$ and $r = 3.85$ agree with the version 1 archive to the stored digits, 1.757509 and 1.863299. The largest absolute gap on the 294 series is zero at the printed precision. The joint-tuple means at those stations are 2.210069 and 2.802431. At every station the corrected means satisfy $0.6 \text{Syn} + 0.4 \text{Surp} = \text{excess3}$.

4 Results

4.1 The short-window maximum is at the grid edge

Figure 1 shows the three scores along the grid. Table 1 lists the stations used below. The largest six-seed mean of excess3 is 2.897 ± 0.006 at $r = 3.95$. Every leave-one-seed maximum falls at that same station. At $r = 3.5675$ the mean is 2.383 ± 0.0004 , and at the neighboring station $r = 3.566168$ it is 2.383 ± 0.0004 . At the nearest grid point to r_∞ the mean is 2.375 ± 0.007 . Across the 33 stations in $[3.544, 3.595]$ every mean lies between 2.344 and 2.383. The band is still a plateau. It sits about 0.51 bits below the grid edge.

At $r = 3.5675$ the two pieces are $\text{Syn} = 0.0137 \pm 0.0021$ and $\text{Surp} = 5.936 \pm 0.004$. Their weighted contributions are 0.0082 and 2.374 bits. Developed chaos at $r = 3.85$ has $\text{excess3} = 2.802 \pm 0.005$, so the accumulation station lies 0.420 bits below it. The Surp term contributes $+0.566$ bits to that difference and the Syn term contributes -0.986 bits. Along the 49 grid means, the correlation of excess3 with Surp is -0.204 , with Syn it is 0.643 , and with the number of distinct tuples it is 0.375 . The largest Syn on the grid is 1.782 ± 0.010 at $r = 3.95$, which is also where excess3 is largest.

4.2 The joint support collapses

Figure 3 places the support next to the block residual. At $r = 3.5675$ a window of 13 symbols contains 4.11 ± 0.02 distinct joint tuples, visited 3.20 times on average, with modal mass 0.304. At $r = 3.30$ the same window contains 11.39 ± 0.05 tuples. At $r = 3.85$ it contains 12.62 ± 0.02 , visited 1.04 times, with modal mass 0.098. The grid maximum, $r = 3.95$, contains 12.79 ± 0.02 tuples.

Figure 2 shows sixteen consecutive symbols from the representative realization at each of those three stations. At $r = 3.5675$ the segment is the cycle of four joint symbols 1, 2, 3, 4, repeated four times. The integers are the order of first appearance in the segment. They label distinct 4-tuples of permutation indices; they are not comparable across panels, and they are not themselves the Bandt–Pompe ranks. At $r = 3.30$ the segment uses eleven joint symbols. At $r = 3.85$ it uses thirteen.

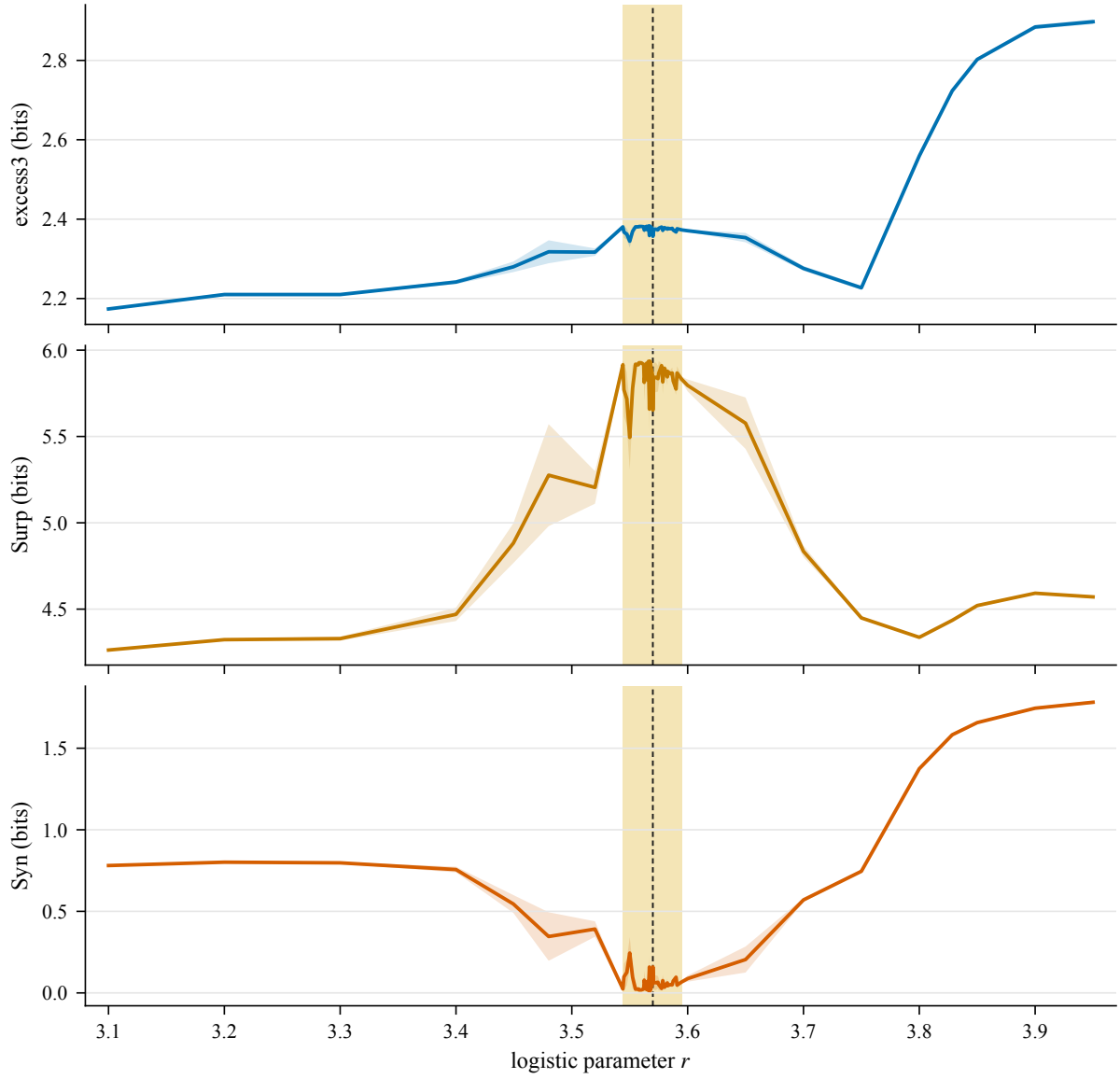


Figure 1: Window-13 scores along the logistic parameter. Curves are six-seed means and shading is the standard error. The band marks $r \in [3.544, 3.595]$. The dashed line is $r_\infty = 3.5699456$. Surp peaks inside the band. excess3 and Syn are largest at the right edge.

Alt text. Three stacked curves against the logistic parameter. Surprise rises to a plateau near 3.57 and then falls. Excess3 and synergy stay lower on that plateau and rise through developed chaos, with their highest point at $r = 3.95$.

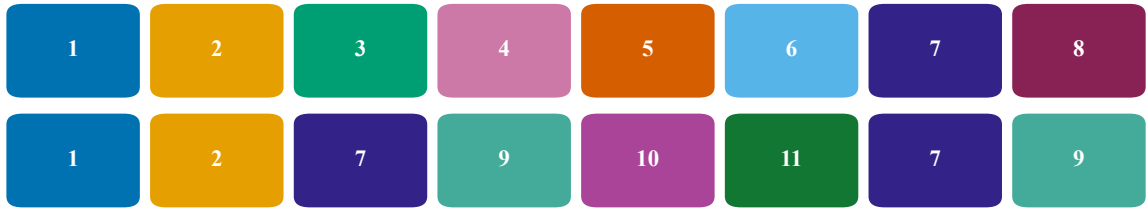
Table 1: Six-seed means at eight stations. The standard error of the mean is in parentheses. Tuples, visits, and modal mass refer to windows of length 13. Res_{pair} is the Kullback–Leibler residual of the full symbol block.

r	regime	excess3	Syn	Surp	tuples	visits	modal	Res_{pair}
3.30000	period 2	2.210 (0.002)	0.7974 (0.0051)	4.329 (0.012)	11.39 (0.05)	1.15	0.147	0.0024 (0.0004)
3.44949	period 4	2.280 (0.013)	0.5454 (0.0546)	4.882 (0.115)	8.91 (0.55)	1.64	0.190	0.0018 (0.0006)
3.54409	period 8	2.381 (0.002)	0.0242 (0.0109)	5.915 (0.021)	4.20 (0.09)	3.16	0.302	0.0005 (0.0002)
3.56617	near r_{∞}	2.383 (0.000)	0.0145 (0.0043)	5.935 (0.007)	4.12 (0.03)	3.20	0.304	0.0005 (0.0003)
3.95000	grid maximum	2.897 (0.006)	1.7817 (0.0096)	4.571 (0.008)	12.79 (0.02)	1.02	0.090	0.1816 (0.0025)
3.56995	r_{∞}	2.375 (0.007)	0.0619 (0.0442)	5.845 (0.083)	4.51 (0.36)	3.04	0.293	0.0031 (0.0028)
3.82843	period-3 window	2.723 (0.007)	1.5823 (0.0094)	4.435 (0.008)	12.56 (0.02)	1.04	0.101	0.1938 (0.0052)
3.85000	developed chaos	2.802 (0.005)	1.6570 (0.0053)	4.521 (0.010)	12.62 (0.02)	1.04	0.098	0.1921 (0.0016)

Alt text. A table of logistic parameter values. Surprise and the tuple count are extreme near $r = 3.5675$, where about four joint tuples appear. Excess3 and synergy are largest at $r = 3.95$. The full-block residual is near zero in the accumulation band and near 0.19 in the period-3 window and in developed chaos.

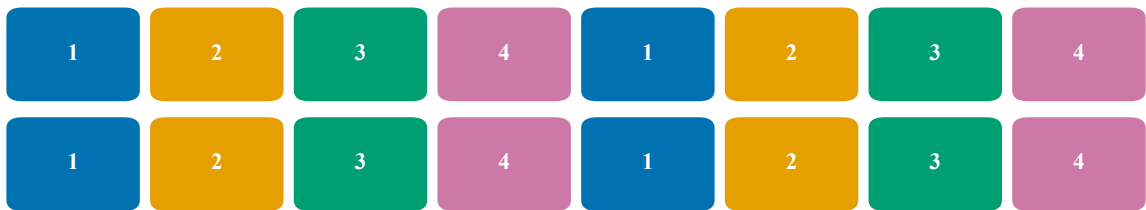
The block residual does not follow the short-window proxy. At $r = 3.5675$ it is 0.0003 ± 0.0001 . At $r = 3.85$ it is 0.192 ± 0.002 , and its grid maximum is 0.198 ± 0.003 at $r = 3.80$. The full-block hybrid is largest in the accumulation band. Its grid maximum is 2.369 at $r = 3.54409$. At $r = 3.85$ the full-block hybrid is 0.316 ± 0.005 , and at $r = 3.95$ it is 0.320 ± 0.005 .

(a) Period 2 $r = 3.30$



11 distinct joint symbols in these 16 steps

(b) Accumulation $r = 3.5675$



4 distinct joint symbols in these 16 steps

(c) Developed chaos $r = 3.85$



13 distinct joint symbols in these 16 steps

Figure 2: Sixteen consecutive joint symbols, read left to right and then onto the second row. The displayed integer is the order of first appearance within the segment. The segment is chosen by the rule in Section 3. (a) Period 2 uses eleven joint symbols. (b) At $r = 3.5675$ the same four joint symbols repeat. (c) Developed chaos uses thirteen joint symbols in the same length.

Alt text. Three pairs of colored rows. The middle pair repeats the colors and numbers 1, 2, 3, 4. The top and bottom pairs introduce many more colors.

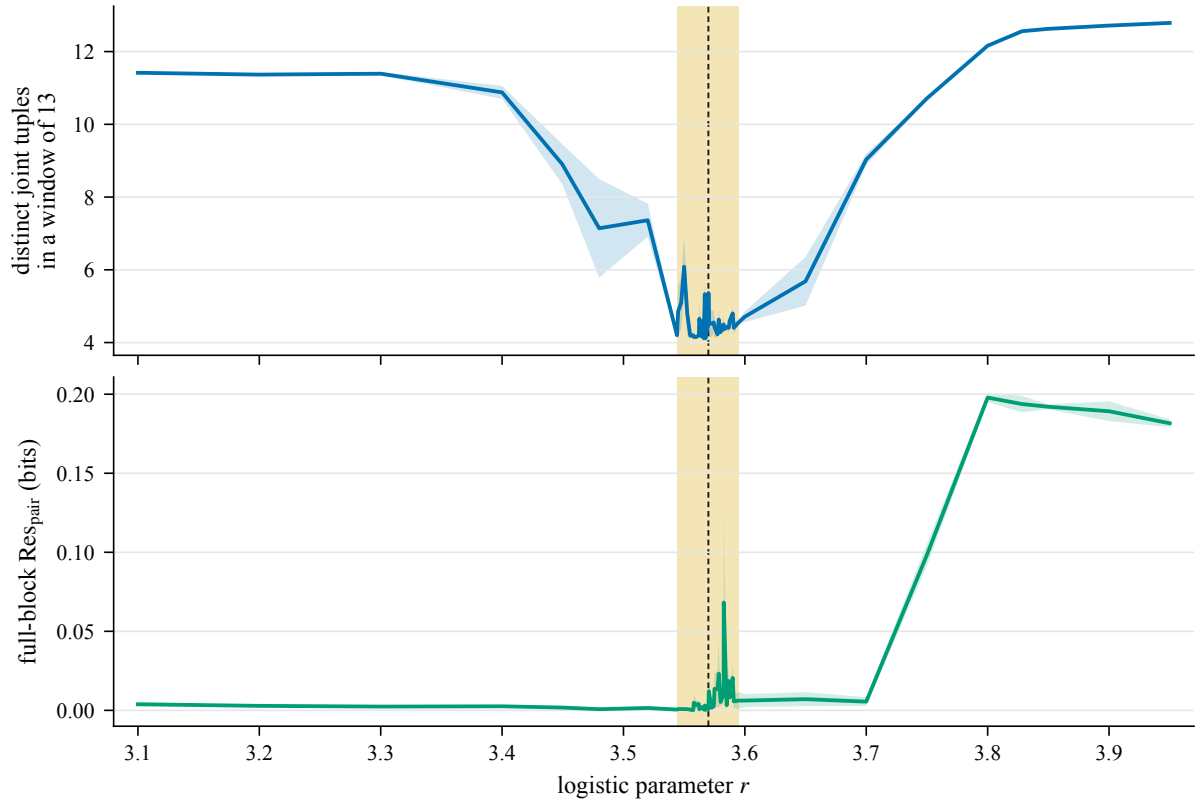


Figure 3: Support and block residual along r . Shading, band, and dashed line are as in Figure 1. The number of distinct joint tuples in a window of 13 is smallest inside the band. The full-block residual stays near zero there, apart from isolated grid points whose standard error is comparable to the mean: at $r = 3.582612$ the residual mean is 0.068 with standard error 0.059. The residual then rises and reaches 0.198 ± 0.003 at $r = 3.80$.

Alt text. Two stacked curves. The count of distinct joint tuples falls to about four inside the yellow band and then rises past twelve. The full-block residual stays near zero through that band and rises to about 0.20 after $r = 3.75$.

4.3 Pairs reproduce the stationary score

Table 2 compares the accumulation station with developed chaos. At $r = 3.5675$, draws from the empirical joint p score 2.161 ± 0.033 , and draws from $P^{(2)}$ score 2.174 ± 0.025 . The ratio of the pairwise score to the sliding score is 0.912. At $r = 3.85$ the same ratio is 0.942, and at the grid maximum $r = 3.95$ the two draw means are 2.743 ± 0.014 and 2.736 ± 0.015 . The pairwise projection accounts for the excess3 of independent draws from the invariant joint at each of these stations. The sliding window exceeds the independent draws from p by 0.221 ± 0.033 bits at $r = 3.5675$. Those windows occupy 4.11 tuples, while the independent draws from the same p occupy 5.54.

Inside single windows the two gaps agree with each other. At $r = 3.5675$ the gap against the window's own pairwise projection is 0.225 ± 0.005 , and the gap against draws from that window's own p is 0.206 ± 0.006 . At $r = 3.85$ the gaps are 0.340 ± 0.010 and 0.325 ± 0.013 . At $r = 3.95$ they are 0.365 ± 0.009 and 0.366 ± 0.008 . The short window does not separate a residual beyond pairs from the resampling fluctuation of a 13-symbol table.

4.4 Long windows return the maximum to the band

Figure 4 extends the hybrid from length 6 to length 312. At $r = 3.5675$ the mean is 2.301 at length 6, 2.383 at length 13, and 2.375 ± 0.008 at length 312. At that last length the accumulation series still occupies only 6.55 joint symbols. At $r = 3.85$ the mean is 2.447 at length 6, 2.802 at length 13, 1.964 at length 104, and 1.135 ± 0.011 at length 312, where the window occupies 199.5 symbols. The argmax of the six-seed curve is $r = 3.95$ for lengths 6 through 52, except length 8, whose argmax is $r = 3.90$. From length 78 through 156 the argmax is $r = 3.566168$. At lengths 208 and 312 it is $r = 3.5575$. Lengthening the window creates the accumulation maximum. It does not preserve the short-window ordering.

The period-3 station $r = 3.828427$ follows the chaotic curve. Its window-13 hybrid is 2.723 ± 0.007 , its support is 12.56 ± 0.02 tuples, its synergy is 1.582 ± 0.009 , and its block residual is 0.194 ± 0.005 . A periodic window is not, by itself, the accumulation plateau.

Period 2 behaves in a third way. By length 312 its joint support has saturated at 32 symbols and excess3 has leveled at 1.248. The period-8 station $r = 3.54409$ is inside the high band of surprise: excess3 = 2.381 ± 0.002 and the short window holds 4.20 ± 0.09 tuples.

Table 2: Accumulation station against developed chaos. Entries are six-seed means $\pm$ the standard error. Independent draws are eighty sequences of length 13. Local gaps use 36 windows and four draws each.

quantity	$r = 3.5675$	$r = 3.85$
excess3, $w = 13$	2.383 ± 0.0004	2.802 ± 0.005
Syn	0.0137 ± 0.0021	1.6570 ± 0.0053
Surp	5.936 ± 0.004	4.521 ± 0.010
distinct tuples, $w = 13$	4.11 ± 0.02	12.62 ± 0.02
visits per tuple	3.20	1.04
modal mass	0.304	0.098
iid draws from P	2.161 ± 0.033	2.629 ± 0.020
iid draws from $P^{(2)}$	2.174 ± 0.025	2.639 ± 0.012
sliding minus iid P	0.221 ± 0.033	0.173 ± 0.021
local gap vs. $P^{(2)}$	0.225 ± 0.005	0.340 ± 0.010
local gap vs. own P	0.206 ± 0.006	0.325 ± 0.013
full-block excess3	2.336 ± 0.054	0.316 ± 0.005
full-block Res _{pair}	0.0003 ± 0.0001	0.1921 ± 0.0016
excess3, $w = 312$	2.375 ± 0.008	1.135 ± 0.011
distinct tuples, $w = 312$	6.55 ± 0.2	199.5 ± 1.2

Alt text. A two-column comparison of $r = 3.5675$ and $r = 3.85$. Surprise and repeated tuples are higher at $r = 3.5675$. The sliding-window score, synergy, and the full-block residual are higher at $r = 3.85$. Independent draws from the joint and from its pairwise projection agree with each other in both columns.

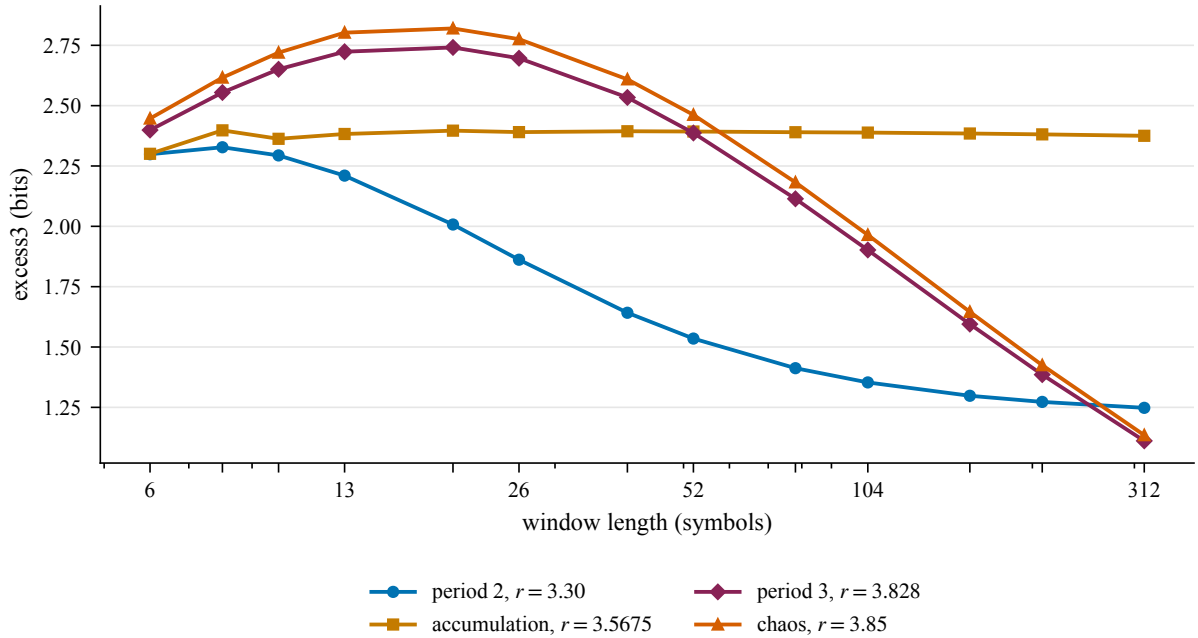


Figure 4: The hybrid against window length at four stations. Points are six-seed means. Lengths above 13 use about 160 evenly spaced windows. The accumulation curve stays near 2.4. The period-3 and chaotic curves rise above it at short lengths and fall toward 1.1 by length 312. Period 2 falls and then levels off.

Alt text. A log-scaled plot of excess3 against window length. The accumulation curve is nearly flat near 2.4. Period 2 declines and flattens near 1.25. Period 3 and developed chaos start higher and fall to about 1.1 at window length 312.

5 Discussion

The support reading survives the recount. Near accumulation the orbit of the joint symbol lives on about four tuples. Those tuples are far from an independent product, so Surp is large, and the full-block Res_{pair} is 0.0003 at the reported precision. Developed chaos widens the support, lowers Surp , and is where the block residual is about 0.19. Synergy, as defined by equation (2), is small on the accumulation plateau and is about 1.7 bits in developed chaos. On windows of length 13 that term sets the location of the hybrid maximum.

A reading that takes the peak of short-window excess_3 as the place where ordinal structure beyond pairs is strongest now points at developed chaos. A reading that takes the collapse of joint support, or the peak of Surp , points at the accumulation band. The two readings separate once Syn is the joint-tuple quantity. They met in version 1 because the flattened count kept Syn below 0.12 everywhere. The long-window hybrid agrees with the support reading: from length 78 the argmax is back inside the band, and the full block is largest at $r = 3.54409$.

Two smaller facts keep the statement inside this ensemble. First, consecutive windows at $r = 3.5675$ are more concentrated than independent draws from the invariant joint, by 0.221 ± 0.033 bits. The local 13-symbol gaps against $P^{(2)}$ and against the window's own p agree, so that concentration is not a resolved residual beyond pairs. Second, one grid point inside the band, $r = 3.582612$, has a block-residual mean of 0.068 with standard error 0.059. Isolated seeds can move a single station. They do not move the six-seed maximum of the residual, which remains at $r = 3.80$.

The scope is the generator in equation (6): four coordinates, coupling 0.05, noise level 0.003, embedding order 3, and the six seeds in equation (7). The weights in equation (4) are the published weights. The paper does not refit them. The result does not derive the operational concordance bands used elsewhere in the framework, and it does not transfer either maximum to other couplings, other dimensions, or series measured in the field.

6 Conclusion

On this coupled logistic cascade, joint-tuple excess_3 on windows of length 13 is largest at $r = 3.95$. The accumulation band is where the joint symbol repeats about four tuples and Surp is largest, and where the full-block residual is quiet. Syn is the term that rises in developed chaos and, at this window length, sets the hybrid maximum. The period-3 window follows that chaotic regime. A longer window moves the hybrid maximum back into the accumulation band.

Author declarations

Conflict of interest. The author has no conflict of interest to disclose.

Ethics approval. Ethics approval is not required. The study uses simulated series only.

Author contribution. Johel Padilla-Villanueva: conceptualization, methodology, software, formal analysis, and writing.

Data availability

This version is archived at <https://doi.org/10.5281/zenodo.22970080>. Version 1 remains <https://doi.org/10.5281/zenodo.22945800>. The generator and the seeds are those of the cascade in Padilla-Villanueva [2026b]. The joint-tuple estimator is nested-recd 0.2.3 (<https://doi.org/10.5281/zenodo.22970079>). The version 1 archive `peak_flat_20260924_145527.csv` and `peak_agg_20260924_145527.csv` is reproduced by `legacy_pooled_counting`. The corrected means are `corrected_flat.csv` and `corrected_agg.csv` from the run of 25 September 2026, together with `mc_stations.json` for the Monte Carlo rows. They accompany this deposit and the repository <https://github.com/johelpadilla/excess3> under `accumulation/data/`. The figures and the numeric tables were drawn by `make_figures.py` from the corrected aggregate joined to the version 1 columns that the bug does not touch.

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